- Classical Statistical Machine Learning setting
  - Functional Risk
  - Empirical Risk
- Gradients
  - Gradient descent algorithm (GD)
  - Stochastic gradient descent (SGD)
- Backpropagation.

General Statistical Machine Learning Setting (Supervised problems)

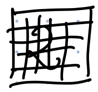
Basic ingredients

1. Space of data $\Omega = \mathbf{X} \times \mathbf{Y}$ — $\mathbf{X}$: examples, $\mathbf{Y}$: labels.

   Usually $X \subseteq \mathbb{R}^d$, $Y \subseteq \mathbb{R}^p$

2. Model, i.e. a function $f : \mathbf{X} \to Y$

   $x \to f(x)$; $x \in X$, $f(x) \in Y = \mathbb{R}^{10}$

   28  28, $\mathbb{R}^{784}$

   $$\begin{array}{c} 0 \\ (1\ 0\ 0\ \dots\ 0) \\ (0\ 1,\ 0\ \quad 0) \\ 1 \end{array}$$

3. Loss function

   $$\ell : \underline{\underline{Y \times Y}} \to \mathbb{R}_+$$

   It is a measure of "closeness" in the lable space

   Ex $\ell(a,b) = \frac{1}{2}\|a-b\|^2$

   $$f(\boxed{4}) = 2$$

   $$(\boxed{2}, 2)$$

   $$(\boxed{2}, 4)$$

4. Probability measure on Ω π

1 + 4 $\Rightarrow (\Omega, \pi)$ is a probability space

Definition (Functional Risk)

Functional: function which has for domain a space of function and it is usuall real-valued

$$F : X \longrightarrow \mathbb{R}$$

Ex. $F(f) = \int_0^1 f$

$$L(f) = \mathbb{E}(\ell \circ (f \times id))$$

$$\ell(\underset{f}{\cdot}, \underset{id}{\cdot}) \quad \cdot \in Y,\ \cdot \in Y$$

$$= \int_\Omega \ell(f(x), y)\, d\pi(x,y)$$

→ "Take a sample (x,y) from Ω apply f to x and compute $f(x)$, Then compare using ℓ $f(x)$ with y."

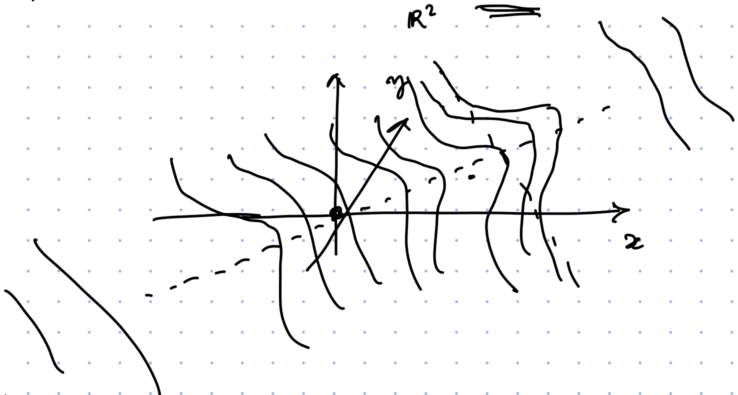
And then sum this quantity over all possible samples weighted with the given probability measure. "

Example 1.

$\Omega = \mathbb{R}^2$

$\rho(x,y) = c_1 \, e^{-(x-y)^2} \, e^{-x^2}$

c_1 is chosen so that $\int_{\mathbb{R}^2} \rho(x,y)\, dx\, dy = 1.$



What is the expected behavior of f

$\underline{f(x) = y}$

Exercise Find

$$\min \{ L(f) : f \in \operatorname{dom}(L) \}$$

using the setting of Example 1. with $\ell(a,b) = \frac{1}{2}(x-y)^2$

$\operatorname{dom}(L)$: is the domain of the functional risk and it is the space of functions for which $\int_{\Omega} \ell(f(x), y)\, d\pi(x,y)$ is finite $(< +\infty)$

Problems

1. We don't know π (the probability measure)
2. Usually we don't have efficient way to solve variational problems.

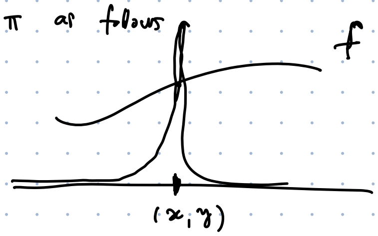
The idea is that since we cannot directly minimize the Functional Risk we generally use a surrogate which is called "Empirical Risk".

Even if we don't have π in many cases it is reasonable to assume to have a i.i.d. samples taken from π.

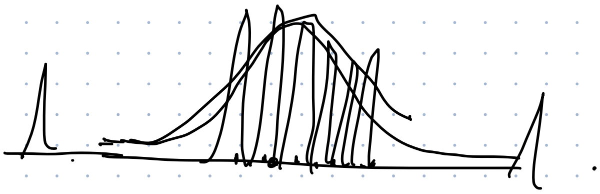
$$D = \{ \underset{\underset{\Omega}{\uparrow}}{(x_1, y_1)}, \dots, \underset{\underset{\Omega}{\uparrow}}{(x_N, y_N)} \}.$$

Then a very direct idea is to approximate π as follows

$$\pi \to \pi_N = \frac{1}{N} \sum_{i=1}^{N} \delta_{(x_i, y_i)} \quad \text{Dirac } \delta.$$

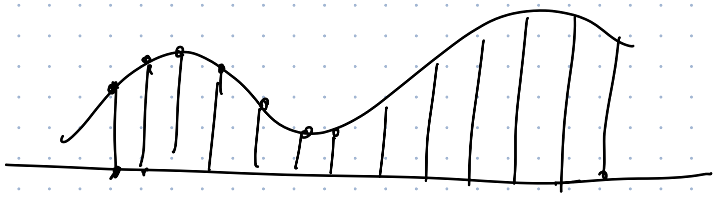


$$\int f \, \delta_{(x,y)} = f(x,y)$$



If we use π_N instead of π we get the Empirical Risk from the functional risk

$$L(f) \to \boxed{L_N(f) = \frac{1}{N} \sum_{i=1}^{N} \ell(f(x_i), y_i)} \quad \text{Empirical Risk}$$



The other simplification that we usually do in ML and in deep learning is to restrict the choice of the model f to a class of models $\mathcal{M}$ that is usually a parametric family of functions.

Ex: $\mathcal{M} = \{ f: \mathbb{R} \to \mathbb{R} : f = ax + b, \ a, b \in \mathbb{R} \}$ (parameters: a, b)

$$f(\cdot\,; \overset{a}{1}, \overset{b}{2}) = x + 2$$

In the next lectures we will choose $\mathcal{M}$ to be Neural Networks.

The real problem that we are going to solve is to choose

$$f^N \in \operatorname{argmin} \{ L_N(f) : f \in \mathcal{M} \}.$$

$$L(f^N) > \inf\{ L(f) : \mathrm{dom}(L) \}$$

$$L(f^N) - \inf\{L(f) : \mathrm{dom}(L)\}$$

$$= L(f^N) \overbrace{- \inf\{L(f) : f \in \mathcal{M}\} + \inf\{L(f) : f \in \mathcal{M}\}}^{=0} - \inf\{L(f) : f \in \mathrm{dom}(L)\}$$

Gradient Descent

if $f \in \mathcal{M}$ then it can be written as $f(\cdot, w)$

w are the parameters that identify the function in $\mathcal{M}$

$w \in \mathbb{R}^n$

Minimizing the ER on $\mathcal{M}$ means

$$\boxed{\min\{L_N(f(\cdot, w)) : w \in \mathbb{R}^n\}}.$$

Now I will call $L_N(f(\cdot, w)) =: F(w)$

Gradient descent:

Fix $w^0 \in \mathbb{R}^n$

Then for $k \geq 0$ compute

$$w^{k+1} = w^k - \tau \nabla F(w^k)$$

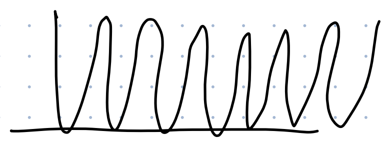
Theorem $F : \mathbb{R}^n \to \mathbb{R}$, $F \in C^1(\mathbb{R}^n; \mathbb{R})$ and assume ∇F is L-Lipschitz

f is L-Lipschitz if $\|f(x) - f(y)\| \leq L \|x - y\|$

$\exists f \in C^1$ this means that ∇f is bounded.

Then assume $\inf F > -\infty$. Then

$$\nabla F(w^k) \to 0 \quad \text{as } k \to +\infty.$$



$f \in C^1$

Proof

$$F(w^{k+1}) = F(w^k - \tau \nabla F(w^k))$$

$x = w^k$

$y = \nabla F(w^k)$

$$f(x - \tau y) = f(x) - \int_0^\tau \nabla f(x - sy) \cdot y \, ds$$

$$\phi(s) = f(x - sy)$$

$$\phi'(s)$$

$$= F(w^k) - \int_0^\tau \nabla F(w^k - s \nabla F(w^k)) \cdot \nabla F(w^k) \, ds.$$

$$= F(w^k) - \int_0^\tau \left(\nabla F(w^k - s\nabla F(w^k)) - \nabla F(w^k) + \nabla F(w^k)\right) \cdot \nabla F(w^k) \, ds$$

$$= F(w^k) - \underbrace{\int_0^\tau \left(\nabla F(w^k - s\nabla F(w^k)) - \nabla F(w^k)\right)\cdot \nabla F(w^k)\, ds}$$

$$- \int_0^\tau |\nabla F(w^k)|^2\, ds \qquad\qquad |a\cdot b| \le \|a\|\,\|b\|$$

$$\le F(w^k) - \tau|\nabla F(w^k)|^2 + \int_0^\tau \left|\underline{\left(\nabla F(w^k - s\nabla F(w^k)) - \nabla F(w^k)\right)\cdot \nabla F(w^k)}\right| ds$$

$$\le F(w^k) - \tau|\nabla F(w^k)|^2 + \int_0^\tau |\nabla F(w^k - s\nabla F(w^k)) - \nabla F(w^k)|\,|\nabla F(w^k)|\, ds$$

$$\le F(w^k) - \tau|\nabla F(w^k)|^2 + L\int_0^\tau |s\nabla F(w^k)|\,|\nabla F(w^k)|\, ds$$

$$= F(w^k) - \tau|\nabla F(w^k)|^2 + L\frac{\tau^2}{2}|\nabla F(w^k)|^2$$

$$= F(w^k) \underbrace{- \tau\left(1 - \frac{\tau}{2}L\right)|\nabla F(w^k)|^2}$$

If $1 - \frac{\tau}{2}L > 0 \qquad \tau < \frac{2}{L}$

$$F(w^n) \le F(w^0) - \sum_{k=0}^{n-1} \tau\left(1 - \frac{\tau}{2}L\right)|\nabla F(w^k)|^2$$

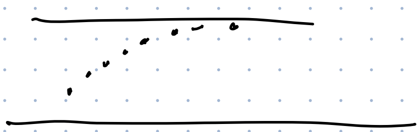
$$F(w^{k+1}) \le F(w^k) - \tau\left(1 - \frac{\tau}{2}L\right)|\nabla F(w^k)|^2$$

$$\le F(w^{k-1}) -$$

$$\inf F \le F(w^k) \le F(w^0) - \sum_{k=0}^{n-1} \tau\left(1 - \frac{\tau}{2}L\right)|\nabla F(w^k)|^2$$

$$\underbrace{\sum_{k=0}^{n-1} \tau\left(1 - \frac{\tau}{2}L\right)|\nabla F(w^k)|^2}_{S_n} \le F(w^0) - \inf(F) \quad \forall n$$

S_n in an increasing sequence uniformly bounded from above



$$\sum_{k=0}^{+\infty} \underbrace{|\nabla F(w^k)|^2} < +\infty$$

Back prop.

$$\nabla F$$

Back prop is an optimal algorithm to compute gradients in Feed forward Neural Networks.

if $\# w = m$

$$\underline{O}(m) \qquad \Theta(m)$$

$$O(m)$$

$$(\nabla F)_i = \frac{F(w_i + \Delta w_i) - F(w_i)}{\Delta w_i}$$

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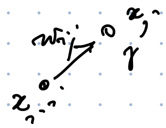
* simple numerical discretization of the gradient for NN

$$O(m^2)$$



$$\frac{\partial \varphi}{\partial w_{ij}} = \underbrace{\delta_i}_{\text{backward step}} \underbrace{x_j}_{} \rightarrow \text{forward pass}$$